

Quiz 2
Multivariable Calculus
BSSE23 - A

Solution:

a) Find area of Δ formed by points A, B & C.

Let's consider $r = 20$

Then

$$3x + y + z = 20$$

$$2x - y + z = 20$$

$$x + 2y - z = 20$$

By solving these simultaneous equations using calculator

$$(x, y, z) = (16, -8, -20)$$

You can also solve the same system using $AX=B$

$$\text{and get } X = A^{-1}B = \begin{bmatrix} 16 \\ -8 \\ -20 \end{bmatrix}$$

$$\text{So, } A = (-20, 0, 0), B = (0, 20, 0), C = (16, -8, -20)$$

$$\text{And Area of } \Delta = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} = (0, 20, 0) - (-20, 0, 0) = \langle 20, 20, 0 \rangle$$

$$\& \vec{AC} = (16, -8, -20) - (-20, 0, 0) = \langle 36, -8, -20 \rangle$$

$$\& \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 20 & 20 & 0 \\ 36 & -8 & -20 \end{vmatrix}$$

$$= (-400 - 0)\hat{i} - j(-400 - 0) + (-720 - 160)\hat{k}$$

$$= -400\hat{i} + 400\hat{j} - 880\hat{k}$$

$$\text{So Area of } \Delta = \frac{1}{2} \sqrt{(-400)^2 + (400)^2 + (-880)^2} = 523.067$$

b) Find parametric equation of the line through Q_0, Q_1 and the point D.

The parametric equation of a line through two points is given by

$$Q(t) = Q_0 + t \vec{Q_0 Q_1} = (0, 0, 0) + t \langle 10, -10, 0 \rangle$$

$$\text{So } Q(t) = \begin{cases} x(t) = 10t \\ y(t) = -10t \\ z(t) = 0 \end{cases}$$

And the line through A & B is

$$P(s) = A_0 + s \vec{AB} = \begin{cases} x(s) = -20 + 20s \\ y(s) = 0 + 20s \\ z(s) = 0 \end{cases}$$

These lines intersect at D, so

$$x(t) = x(s) \quad \text{and} \quad y(t) = y(s)$$

$$10t = -20 + 20s \quad -10t = 20s$$

$$10t = -20 + 20(-t/2) \quad s = -t/2$$

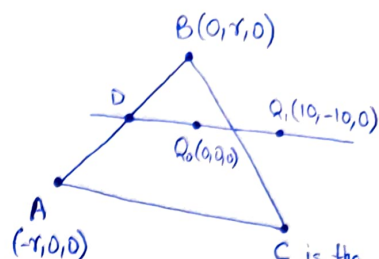
$$10t = -20 - 10t$$

$$\boxed{t = -1} \quad \& \quad \boxed{s = \frac{1}{2}}$$

by plugging $t = -1$ in $Q(t)$, we get

$$\text{so } D = (-10, 10, 0)$$

! we get same result by plugging s in $P(s)$.



C is the intersection of planes

$$\begin{cases} 3x + y + z = r \\ 2x - y + z = r \\ x + 2y - z = r \end{cases}$$